FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE — GRADE 10 | SUB-STRAND 2.5: RATES OF REACTIONS	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent eight lessons investigating why some chemical reactions happen in a flash while others take days. Now it is your turn to bring everything together and answer the driving question using evidence, particle-level thinking, and real Kenyan examples.

READ THIS FIRST — What makes a great Final Explanation?
A strong Final Explanation does three things:
1. It answers the driving question directly and completely.
2. It uses evidence from investigations (the thiosulfate cross, experiments on temperature, concentration, surface area, catalysts, and light) to support every claim.
3. It connects particle-level reasoning to real-life situations in Kenya.

YOUR DRIVING QUESTION:
"Why do some chemical reactions happen in a flash while others take days — and how can we control the speed of reactions to make life in Kenya safer, healthier, and more productive?"

INSTRUCTIONS:
- Write your answer in the four sections below. Each section has a guiding prompt and a model sentence to help you get started.
- Use scientific vocabulary correctly: collision theory, activation energy, reaction rate, concentration, surface area, catalyst, temperature, frequency of collisions, successful collisions.
- Reference at least THREE real Kenyan contexts (e.g., cooking githeri, fermenting uji, rusting of iron roofing sheets, enzyme activity, maize flour milling, tea processing in Kericho).
- You may draw and annotate particle diagrams to support your written explanations — diagrams count as evidence.
- Your explanation should be detailed enough that a fellow Grade 10 student who missed all eight lessons could read it and fully understand rates of reactions.

Section 1 — The Core Idea: What IS a Reaction Rate and Why Does It Vary?

Explain what 'rate of reaction'	The rate of a reaction tells us how quickly reactants are converted into products — in our case, how fast the thiosulfate and

means at the particle level. Use the Disappearing Thiosulfate Cross phenomenon as your starting point. Why did the cross disappear faster in Beaker B (warm, ~50 °C) than in Beaker A (cold, ~10 °C), even though both beakers contained exactly the same chemicals in the same amounts? In your answer, describe what particles are doing inside the beaker and what must happen for a reaction to actually occur.	hydrochloric acid produced enough sulfur precipitate to block the view of the cross. According to collision theory, chemical reactions only happen when reactant particles collide with sufficient energy and in the correct orientation — these are called successful collisions. In Beaker A (cold water, ~10 °C, like a cool Nairobi highland morning), particles move slowly, so they collide less frequently and with less energy. Very few collisions meet the activation energy threshold, so sulfur forms slowly and the cross stays visible for several minutes. In Beaker B (warm water, ~50 °C, like a hot afternoon in Mombasa), particles move much faster. They collide more often AND with greater energy, so a much higher proportion of collisions are successful. The sulfur precipitate forms rapidly — the cross disappears in about 30–40 seconds. Both beakers contained identical chemicals, so the only variable was the kinetic energy of the particles. This is the foundation of everything: reaction rate depends on the frequency and energy of particle collisions.

Section 2 — The Factors: How Can We Speed Up or Slow Down a Reaction?

Describe FOUR factors that affect the rate of a chemical reaction. For each factor, explain (a) what changes at the particle level, (b) how it increases or decreases the rate, and (c) one specific Kenyan example that shows this factor at work in everyday life. You must include: temperature, concentration, surface area, and catalysts.	Four key factors control how fast a reaction occurs: 1. TEMPERATURE — Increasing temperature gives particles more kinetic energy, so they move faster, collide more frequently, and more collisions exceed the activation energy. This is why githeri (a mixture of maize and beans) cooks faster when the water is at a rolling boil than when it barely simmers. It is also why food stored in a cool Kericho highland evening stays fresh longer than food left out in the Mombasa afternoon heat — lower temperatures slow the reactions of decay. 2. CONCENTRATION — A higher concentration means more reactant particles are packed into the same volume, so collisions happen more often. When a farmer in the Rift Valley uses a more concentrated pesticide solution, it reacts more rapidly with target pests. In our thiosulfate experiment, if we had used a more concentrated sodium thiosulfate solution, the cross would have disappeared even faster because sulfur-producing collisions would have been more frequent. 3. SURFACE AREA — Breaking a solid into smaller pieces exposes more particle surfaces to the reactant, increasing the number of collisions per second. When maize is ground into fine unga (flour) at a posho mill in Kisumu, its surface area increases enormously compared to whole maize kernels — this is why flour dissolves and reacts in cooking much faster than whole grain. Similarly, finely divided charcoal ignites faster than a large log. 4. CATALYSTS — A catalyst is a substance that increases the rate of a reaction without being used up itself. It works by providing an alternative reaction pathway with a lower activation energy, so more collisions are successful even without adding heat. Enzymes are biological catalysts. In our own bodies, the enzyme amylase in saliva begins breaking down the starch in ugali the moment we start chewing, making digestion faster and more efficient. Industrially, catalysts are used in fertiliser production at factories that supply Kenyan farmers.

Section 3 — Evidence From Our Investigations: Connecting Data to Claims

Choose TWO of the investigations you carried out during this unit. For each one, describe: (a) what you did, (b) the pattern you observed in the results (refer to specific observations or trends, not just 'it got faster'), and (c) how the data supports your explanation of reaction rates at the particle level. If you drew graphs or recorded times, describe those results here.	Investigation 1 — Temperature and the Thiosulfate Cross: We set up beakers of sodium thiosulfate and hydrochloric acid at different temperatures (approximately 10 °C, 25 °C, 40 °C, and 50 °C) and measured the time taken for the cross to disappear. Our results showed a clear pattern: as temperature increased, the time taken decreased sharply. For example, at 10 °C the cross took over 3 minutes to disappear, but at 50 °C it disappeared in under 40 seconds. When we plotted reaction rate (1/time) against temperature, the graph showed a steep upward curve. This supports collision theory: at higher temperatures, particles have more kinetic energy, move faster, and produce more frequent, more energetic collisions — so the rate of sulfur formation increases rapidly with temperature. Investigation 2 — Concentration: We kept temperature constant but varied the concentration of sodium thiosulfate solution, using dilutions of 100%, 75%, 50%, and 25% of the original concentration. We again measured the time for the cross to disappear. The results showed that the most concentrated solution produced the fastest reaction — the cross disappeared quickest — while the most dilute solution reacted slowest. A graph of rate (1/time) against concentration showed a positive, roughly linear relationship. This is consistent with particle theory: in a more concentrated solution, more thiosulfate particles occupy the same volume, so the frequency of collisions between thiosulfate and HCl particles is greater, producing more successful collisions per second and therefore a faster rate of sulfur precipitation.

Section 4 — Applying Your Knowledge: Making Life in Kenya Safer, Healthier, and More Productive	
Now answer the second half of the driving question. Using what you know about factors that affect reaction rates, explain THREE specific ways that controlling reaction speed can make life in Kenya safer, healthier, or more productive. For each example, clearly state WHICH factor is being used and HOW it is being applied. Think beyond the classroom — consider farmers, food producers, health workers, engineers, and everyday Kenyan families.	Understanding and controlling reaction rates has real, practical value for people across Kenya: 1. SAFER — Slowing Rusting of Iron Roofing Sheets (Temperature & Surface Area): Millions of Kenyan homes and shops are roofed with iron sheets (mabati). Iron reacts with oxygen and water in a slow oxidation reaction that we call rusting. This reaction is accelerated by heat and moisture, which is why roofs in humid coastal Mombasa rust faster than those in the cool highlands of Nyahururu. To slow this reaction, engineers coat iron sheets with a layer of zinc (galvanisation) — this acts as a barrier that reduces the surface area of iron exposed to oxygen and water, dramatically slowing the rate of the rusting reaction. Families also paint iron structures to achieve the same protective effect. Slowing this reaction means roofs and structures last longer, saving money and keeping buildings safe. 2. HEALTHIER — Controlling Fermentation of Uji and Preserving Food (Temperature): Uji (fermented porridge made from millet or sorghum flour) is a nutritious traditional food, especially for young children. Fermentation is a chemical reaction carried out by microorganisms. If uji is left in a warm kitchen overnight, the fermentation reaction speeds up, souring the porridge quickly and potentially allowing harmful bacteria to grow. Kenyan mothers and food vendors manage this by storing uji and other fermented foods in cool conditions — lowering the temperature reduces the kinetic energy of the microbial enzymes, slowing the fermentation rate and keeping food safe for longer. The same principle explains why refrigerating sukuma wiki, tomatoes, and nyama choma extends their freshness: cold temperatures slow the biochemical reactions of spoilage. 3. MORE PRODUCTIVE — Speeding Up Digestion and Agricultural Processes (Catalysts & Surface Area): In Kenya's tea industry in Kericho, tea leaves are broken and rolled during processing to increase their surface area — this speeds up the oxidation reactions that develop the characteristic flavour and colour of black tea. On a biological level, the enzymes in our digestive system act as catalysts that allow the complex molecules in foods like ugali, beans, and meat to be broken down quickly at normal body temperature (37 °C). Without these biological catalysts, the same reactions would require far more energy and

	would happen far too slowly to support life. Agricultural researchers at institutions like KARI (Kenya Agricultural and Livestock Research Organisation) also study enzyme catalysts in soil to help farmers improve nutrient availability for crops, increasing yields and food security across the country.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Collision Theory & Particle-Level Explanation	Consistently and accurately explains reaction rate using collision theory throughout all sections. Clearly distinguishes between frequency of collisions and successful collisions (those meeting activation energy). Particle-level reasoning is precise, consistent, and directly tied to the thiosulfate phenomenon and all factors discussed.	Correctly uses collision theory in most sections. Mentions both collision frequency and activation energy/energy of collisions, though the distinction may not always be explicitly stated. Particle-level reasoning is mostly accurate with only minor gaps or imprecision.	Attempts to use collision theory but explanations are incomplete, vague, or contain misconceptions (e.g., states 'particles move faster' without connecting this to collision frequency or successful collisions). Particle-level reasoning is limited to one or two factors or is inconsistently applied.
Coverage and Accuracy of the Four Factors	All four factors (temperature, concentration, surface area, catalysts) are explained accurately and in detail. For each factor, the student clearly describes the particle-level mechanism, the direction of effect on rate, and a relevant, correctly applied Kenyan example. No factual errors are present.	All four factors are addressed and are mostly accurate. Particle-level mechanisms are described for most factors. At least two Kenyan examples are correctly applied. Minor inaccuracies or missing detail in one factor do not significantly undermine the overall explanation.	Fewer than four factors are covered, OR one or more factors contain significant factual errors (e.g., confusing concentration and surface area effects, or stating that a catalyst is 'used up' in the reaction). Kenyan examples are absent, incorrect, or not connected to the relevant factor.
Use of Evidence from Investigations	Two investigations are described with specific, accurate reference to observed data trends (e.g., time values, graph shapes, direction of change). Evidence is explicitly and correctly linked to particle-level claims. The student clearly distinguishes between what was observed and the explanation of why it occurred.	Two investigations are described and evidence is linked to claims, but data references may be general rather than specific (e.g., 'the reaction was faster' without quantitative detail). The connection between evidence and particle theory is present but may lack precision in one area.	Only one investigation is described, OR descriptions are vague and not tied to specific observations or data trends. The student recounts what was done (procedure) but does not clearly connect results to particle-level explanations or reaction rate claims.
Real-Life Application to Kenyan Contexts	Three distinct, accurate real-life Kenyan applications are explained. Each application clearly identifies which factor is being used, how it is being controlled (increased or decreased), and the benefit to Kenyan safety, health, or productivity. Examples draw authentically on Kenyan food, agriculture, industry, or health contexts.	At least two Kenyan applications are explained with the relevant factor identified and a clear benefit stated. A third application may be present but less fully developed. Examples are appropriate and Kenyan in context, though one may be generic rather than specifically localised.	Fewer than two Kenyan applications are provided, OR examples are present but do not identify the specific factor being controlled, OR the applications are not genuinely Kenyan in context (e.g., uses generic Western examples with a Kenyan name added superficially).

Scientific Vocabulary and Communication	Uses scientific vocabulary accurately and consistently throughout: reaction rate, collision theory, activation energy, concentration, surface area, catalyst, kinetic energy, successful collisions, frequency of collisions. Writing is clear, well-organised, and logically sequenced. A fellow student with no prior knowledge could follow the explanation from start to finish.	Most key vocabulary terms are used correctly. Occasional imprecise word choice does not obscure meaning. The explanation is generally well-organised and readable, though one or two sections may lack smooth logical flow.	Scientific vocabulary is limited, missing, or used incorrectly in multiple places (e.g., using 'speed' of reaction without defining it, or misusing 'catalyst' to mean any substance that speeds things up). The explanation is difficult to follow in places, and a reader unfamiliar with the topic would struggle to understand the argument.